

find a way to keep salt out of the now-pure water.

Water from air

With the impact of climate change making ground and surface water more scarce, companies are looking to tap the atmosphere that holds 37.5 million billion gallons of water, calling it water from air. This is higher than the approximately 32.6 million billion gallons of fresh water—the rest is locked up in ice-caps and glaciers—present on earth.

Two primary techniques to extract water from air are based on cooling and use of desiccants. Pulling water out of the atmosphere requires significant input of energy; however, some atmospheric water-generation methods are completely passive, relying on natural temperature differences and requiring no external energy source.

Most atmospheric water generators that make water from air operate in a manner similar to a dehumidifier. In a dehumidifier, air is passed over

a cooled coil, causing the water to condensate. Rate of water generation depends on ambient conditions of temperature and humidity, along with volume of air being passed over the coil as well as capacity of the machine to cool the coil.

As a consequence, such systems reduce air temperature, which, in turn, reduces air's capacity to carry water, lowering water production. Atmospheric water generators are designed for towns, villages, factories, off-grid settlements and communities. These can also be used to provide drinking water in community centres and public places like healthcare centres, schools, colleges, offices, railway stations, bus stands, airports, seaports, off-shore oil rigs, military establishments, remote field areas and establishments, and residential complexes.

While the concept is not new in the US, Europe and the Middle East, the technology—which works well in coastal areas where humidity is high—is yet to achieve scale in India.

Along with companies like Watergen (an Israeli company), progressing well in generating water from air, Mumbai-based WaterMaker (India) Pvt Ltd, Kolkata-based AKVO, Hyderabad-based Skywater Pvt Ltd and Bharat Electronics Ltd (BEL) have also developed different models of atmospheric water generators with the purpose to condense moisture in the atmosphere into water.

Water is then filtered and purified through several filters, including carbon, reverse osmosis and UV sterilisation lights, to produce pure drinking water. It is expected that the magic of being able to pull water out of thin air will soon become a reality. Watergen has recently unveiled a water generator for the average consumer using a simple plug-and-play technique, which will turn air into water.

In the end, it is worth mentioning that electricity and fossil fuels are not the only ways to power atmospheric water generators; solar power is another energy-efficient option. **EFY**

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